

MATH GU4053 INTRO TO ALGEBRAIC TOPOLOGY
HOMEWORK 3

DUE: WEDNESDAY, FEBRUARY 18, 2026, **BEFORE 23:59PM**

SUBMIT THE COMPLETED ASSIGNMENT TO <https://www.gradescope.com>

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- This assignment has **5 questions** on **two pages**.
 - This assignment counts towards 2% of your final grade (up to a maximum of 10% for all homework).
 - Your solutions must be *handwritten*. For instance on a tablet/iPad, or with pen and paper scanned to a pdf.
 - While collaborative work is allowed and generally beneficial, you must write up the solutions *on your own* and *in your own words*.
 - Rigor, clarity, and proper use of mathematical notation count! The work you submit for evaluation will require showing all steps and a clear explanation of what you have done. The explanations need not be lengthy to be clear.
 - Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)
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1. Let $G \times X \rightarrow X$ be a covering space action. Assume that X is simply connected, and let $x \in X$ be a base point. In class we have constructed an isomorphism

$$\phi_G : G \rightarrow \pi_1(X/G, Gx),$$

given by $\phi_G(g) = [q \circ \rho_g]$, where $q : X \rightarrow X/G$ is the quotient map, and $\rho_g : I \rightarrow X$ is any path from x to $g.x$. (In class we wrote just ϕ instead of ϕ_G .)

- (a) Let $H < G$. Prove that the action $H \times X \rightarrow X$, obtained by restricting the action of G , is also a covering space action.

We therefore get an isomorphism

$$\phi_H : H \rightarrow \pi_1(X/H, Hx).$$

- (b) Let $f : X/H \rightarrow X/G$ be the map defined by $f(Hx) = Gx$. Prove that the induced homomorphism

$$f_* : \pi_1(X/H, Hx) \rightarrow \pi_1(X/G, Gx)$$

satisfies $(f_* \circ \phi_H)(h) = \phi_G(h)$ for all $h \in H$.

2. A *manifold* of dimension d is a topological space X that is Hausdorff, and such that any point $x \in X$ has a neighborhood $U \subset X$ which is homeomorphic to $\mathbb{R}^d$.

Assume X is a manifold of dimension $d \geq 2$, let $x_0, x_1 \in X$ be distinct points, and let $i : X \setminus \{x_1\} \rightarrow X$ denote the inclusion map. Prove that the induced homomorphism

$$i_* : \pi_1(X \setminus \{x_1\}, x_0) \rightarrow \pi_1(X, x_0)$$

is surjective.

3. Let Y be a topological space and let $\text{Homeo}(Y)$ denote the set of homeomorphisms $Y \rightarrow Y$, made into a group with respect to composition.

(i) Let G be a group and $\phi : G \rightarrow \text{Homeo}(Y)$ be a group homomorphism. Prove that

$$\begin{aligned} G \times Y &\rightarrow Y \\ (g, y) &\mapsto \phi(g)(y) \end{aligned}$$

defines a continuous action of G on Y .

(ii) Conversely, prove that any continuous action of G on Y arises in this way, for a uniquely determined group homomorphism $\phi : G \rightarrow \text{Homeo}(Y)$.

4. Let $R, S, T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the homeomorphisms defined by

$$\begin{aligned} R(x, y, z) &= (x + 1, y, z + y) \\ S(x, y, z) &= (x, y + 1, z) \\ T(x, y, z) &= (x, y, z + 1). \end{aligned}$$

(a) Prove that $T \circ R = R \circ T$, $T \circ S = S \circ T$, and $S \circ R = R \circ S \circ T^{-1}$. Explain briefly why this proves that the set

$$G = \{R^a \circ S^b \circ T^c \mid (a, b, c) \in \mathbb{Z}^3\}$$

forms a group with respect to composition.

(b) Prove that the function

$$f : \mathbb{Z}^3 \rightarrow G,$$

defined by $f((a, b, c)) = R^a \circ S^b \circ T^c$, is a bijection. Deduce that G is not commutative.

(c) Prove that the action $G \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $g.v = g(v)$ is a covering space action.

5. Let $n \in \mathbb{Z}$. A topological space X is called *n-connected* if for any $k \in \{0, \dots, n + 1\}$ and any map $f : S^{k-1} \rightarrow X$, there exists a continuous map $F : D^k \rightarrow X$ with $F|_{S^{k-1}} = f$.

Prove that X is 1-connected if and only if X is simply connected.

Remark (not needed for doing the problem): Compare this notion with “weakly contractible” from HW1. By the arguments from that problem, X is 0-connected if and only if it is path connected, (-1) -connected if and only if $X \neq \emptyset$, and any X is (-2) -connected.